

Module 14: Unit 5 AKTU Exam PYQs & Numerical Bank

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UNIT 5: AKTU SEMESTER EXAM PYQS & NUMERICAL BANK

Gateway Classes Past 8 Years Solved Trends (2016-17 to 2023-24) | High-Yield Answer Blueprints

Top High-Frequency AKTU Theory Questions

- 1. TELNET Working & Procedure (4 Times Repeated!)**
AKTU 2016-17, 2017-18, 2018-19, 2023-24: Port 23, NVT character set, IAC.
- 2. DNS Divisions & Resolution (AKTU 2018-19, 2021-22)**
Generic, Country, Inverse domains + Recursive vs Iterative resolution.
- 3. HTTP vs HTTPS & Methods (AKTU 2021-22, 2022-23)**
Port 80 vs 443, SSL/TLS security layer, GET vs POST vs HEAD.
- 4. FTP vs TFTP vs HTTP Comparison (AKTU 2023-24)**
TCP Ports 20/21 vs UDP Port 69 vs TCP Port 80, active vs passive mode.
- 5. SNMP Framework (AKTU 2018-19, 2022-23, 2023-24)**
SMI, MIB, SNMP roles, UDP 161/162, 3 management functions.

Solved Numerical & Protocol Tracing Models

- 1. RSA Character "F" Transmission (AKTU 2022-23)**
 $p=3, q=11, n=33, \phi(n)=20, e=3, d=7, P=5$
Encrypt: $C = 5^3 \bmod 33 = 26$ | Decrypt: $P = 26^7 \bmod 33 = 5$ ("F")
- 2. Caesar Cipher "HELLO" Shift (Slide 75)**
 $C = (P + 3) \bmod 26 \rightarrow H \rightarrow K, E \rightarrow H, L \rightarrow O, L \rightarrow O, O \rightarrow R \rightarrow "KHOOR"$
- 3. Run-Length Encoding Compression Ratio**
Raw: 12 Bytes $\rightarrow$ Compressed: 10 Bytes (CR = 1.2 : 1)
- 4. Passive FTP Port Math**
227 Entering Passive Mode (192,168,1,10, 8, 4) $\rightarrow$ Port = $8 \times 256 + 4 = 2052$

AKTU 10-MARK ANSWER WRITING BLUEPRINT: APPLICATION LAYER

1. Protocol Diagram Mandate

- Always draw client-server interaction diagrams.
- Clearly mark Port numbers: DNS (53), HTTP (80), TELNET (23), SMTP (25), POP3 (110).
- State whether protocol is PUSH or PULL.

2. Mathematical Derivations

- In RSA, explicitly state Euler totient formula $\phi(n) = (p-1)(q-1)$ before computing d .
- Write Extended Euclidean tabular steps.
- Verify decryption step explicitly: $C^d \bmod n$.

3. Comparative Questions

- Use tabular formats for comparison questions:
 - HTTP vs HTTPS | POP3 vs IMAP4
 - FTP vs TFTP | Symmetric vs Asymmetric
- Mention exact RFC standard numbers.

1. High-Frequency Exam Blueprint (AKTU 2016-17 to 2023-24)

Comprehensive solved question repository mapping all repetitive theoretical inquiries and quantitative problems from past university semester examinations.

The Top 5 High-Yield Questions

- TELNET Architecture & NVT Working Procedure (Repeated 4 times: 2016, 2017, 2018, 2023).
- DNS Hierarchy, Divisions & Recursive vs Iterative Resolution.
- Master Comparisons: HTTP vs HTTPS, FTP vs TFTP, POP3 vs IMAP4.
- SNMP Triad: SMI, MIB, SNMP & Three Management Functions.
- RSA Key Generation Derivation and Numerical Computation (Character 'F').

2. Solved Numerical Problem Review

Problem Type	Input Parameters	Mathematical Output
RSA Character "F" Transmission	$p=3, q=11, e=3, P=5$	$n=33, \phi(n)=20, d=7, C=26$, Decrypted $P=5$ ("F")
Passive FTP Data Port Extraction	227 Passive (192,168,1,50, 16, 32)	Port = $16 \times 256 + 32 = 4128$
Caesar Cipher Monoalphabetic Shift	Plaintext = "HELLO", Shift $k = 3$	Ciphertext = "KHOOR"